



People's Empowerment Group
ISBM College of Engineering, Nande, Pune
Department of Artificial Intelligence and Machine Learning
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Project Title:

AI – Powered Employee Attrition Prediction and Workforce Intelligence System

Group Members:

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Problem Definition:

Employee attrition is a major challenge for organizations as it increases recruitment costs, reduces productivity, and affects business performance. Existing methods often fail to identify employees who are likely to leave in advance. This project uses AI to predict employee attrition, explain the key reasons behind the prediction, and provide smart recommendations to help HR make better decisions and improve employee retention.

Project Domain:

Artificial Intelligence and Machine Learning (AIML) with Human Resource (HR) Analytics

Literature Survey:

Sr. No.	Title of the Paper	Author / Publication / Year	Technique Used	Summary
1.	Employee Attrition Prediction Using Bayesian Optimized Stacked Ensemble Learning and Explainable AI	Karabi Konar et al. / <i>SN Computer Science (Springer)</i> / 2025	- Bayesian Optimization - Stacked Ensemble - SHAP	Proposes an optimized ensemble learning model with Explainable AI to improve attrition prediction accuracy and interpretability for HR managers.
2.	HR Analytics: Predicting Employee Turnover using Machine Learning Models	<i>IEEE ICDISS Conference</i> / 2025	- Random Forest - Gradient Boosting - Logistic Regression	Compares multiple ML models for employee turnover prediction and demonstrates how predictive analytics supports proactive HR decision-making.
3.	Mitigating Attrition: Data-Driven Approach Using Machine Learning and Data Engineering	Naveen Edapurath Vijayan / <i>arXiv</i> / 2025	- Machine Learning - Feature Engineering - SHAP	Presents a data engineering and machine learning framework for predicting attrition with interpretable insights to support retention strategies.
4.	Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics	Maytha Al-Ali et al. / <i>Scientific Reports (Nature)</i> / 2026	- Adaptive Boosting - Histogram Gradient Boosting - SHAP	Integrates advanced ML with Explainable AI to predict employee attrition and identify the most influential factors affecting employee turnover.
5.	Explainable AI for Employee Turnover Prediction: A SHAP-Based Intelligent Analytics Approach	Haojie Jia / <i>Scientific Reports (Nature)</i> / 2026	- SHAP - Explainable AI	Focuses on transparent employee turnover prediction using SHAP while considering fairness, threshold calibration, and responsible AI practices.
6.	Explainable Attrition Risk Scoring for Managerial Retention Decisions in Human Resource Analytics	M. S. Pavithran & S. M. Vadivel / <i>Frontiers in Big Data</i> / 2026	- Explainable AI - Predictive Analytics - Risk Scoring	Develops an explainable attrition risk scoring system to help managers identify at-risk employees and improve retention decisions.

Scope of the Project:

- Predict employee attrition using AI and machine learning techniques.
- Identify the key factors influencing employee attrition through Explainable AI.
- Provide smart retention recommendations to help HR reduce employee turnover.
- Generate manager decision summaries for better workforce planning.
- Visualize HR insights using an interactive analytics dashboard.
- Support data-driven HR decision-making to improve employee retention and organizational performance.

Software Requirements:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook / Visual Studio Code
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow/Keras, SHAP, Plotly, Matplotlib, Streamlit
- **Database:** SQLite
- **Version Control:** Git & GitHub

Hardware Requirements:

- **Processor:** Intel Core i5 or AMD Ryzen 5 (or higher)
- **RAM:** 8 GB minimum (16 GB recommended)
- **Storage:** 10 GB free disk space
- **Internet Connection:** Required for package installation and LLM/API integration

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Project Guide

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